# Recitation 6

Correlation, Independence, and Causation

Solutions and Teaching Notes

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October 16, 2026

*Reading:* *Probability and Statistics for Data Science*, Sections 8.2–8.6 and 8.8.

## 1 Review: covariance and correlation

Assume finite second moments. Write $\mu_{\widetilde{x}} = \mathbb{E}[\widetilde{x}]$ and $\sigma_{\widetilde{x}}^2 = \mathrm{Var}(\widetilde{x})$.

$$\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = \mathbb{E}[(\widetilde{x} - \mu_{\widetilde{x}})(\widetilde{y} - \mu_{\widetilde{y}})] = \mathbb{E}[\widetilde{x}\widetilde{y}] - \mu_{\widetilde{x}}\mu_{\widetilde{y}}.$$

If both variances are positive, the **correlation coefficient** is

$$\boxed{\rho_{\widetilde{x},\widetilde{y}} = \frac{\mathrm{Cov}(\widetilde{x}, \widetilde{y})}{\sigma_{\widetilde{x}}\sigma_{\widetilde{y}}}.}$$

Correlation measures the direction and strength of a *linear* association. It is dimensionless; covariance depends on the units.

**Properties to remember.**

- **Symmetry and bounds:** $\rho_{\widetilde{x},\widetilde{y}} = \rho_{\widetilde{y},\widetilde{x}}$ and $-1 \le \rho_{\widetilde{x},\widetilde{y}} \le 1$.
- **Shifts and rescaling:** for nonzero constants $b, d$,

  $$\mathrm{Cov}(b\widetilde{x} + a, d\widetilde{y} + c) = bd\ \mathrm{Cov}(\widetilde{x}, \widetilde{y}), \qquad \rho_{b\widetilde{x}+a,d\widetilde{y}+c} = \mathrm{sign}(bd)\rho_{\widetilde{x},\widetilde{y}}.$$

  Positive changes of scale leave correlation unchanged; reversing one variable's sign reverses the correlation.
- **Perfect correlation:** $|\rho| = 1$ exactly when $\widetilde{y} = \alpha + \beta\widetilde{x}$ with probability one, for some $\beta \neq 0$. The sign of $\rho$ is the sign of $\beta$.
- **Variance of a sum:** $\mathrm{Var}(\widetilde{x} + \widetilde{y}) = \mathrm{Var}(\widetilde{x}) + \mathrm{Var}(\widetilde{y}) + 2\,\mathrm{Cov}(\widetilde{x}, \widetilde{y})$.

**Connection to linear prediction.** The best *affine* predictor of $\widetilde{y}$ from $\widetilde{x} = x$ is

$$\ell_{\mathrm{MMSE}}(x) = \mu_{\widetilde{y}} + \frac{\mathrm{Cov}(\widetilde{x}, \widetilde{y})}{\mathrm{Var}(\widetilde{x})}(x - \mu_{\widetilde{x}}) = \mu_{\widetilde{y}} + \rho_{\widetilde{x},\widetilde{y}}\frac{\sigma_{\widetilde{y}}}{\sigma_{\widetilde{x}}}(x - \mu_{\widetilde{x}}).$$

Its MSE is $(1 - \rho_{\widetilde{x},\widetilde{y}}^2)\sigma_{\widetilde{y}}^2$. Thus $\rho^2$ is the fraction of variance explained by the best affine predictor. If $\rho = 0$, that predictor is just the mean; nonlinear prediction may still help. A constant variable has zero covariance with every variable, but its correlation coefficient is *undefined*.

## 2 Mini-lecture: uncorrelated versus independent

**Uncorrelated** means $\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0$. **Independent** means that, for every pair of events defined by the two variables,

$$\mathbb{P}(\widetilde{x} \in A, \widetilde{y} \in B) = \mathbb{P}(\widetilde{x} \in A)\mathbb{P}(\widetilde{y} \in B).$$

For discrete variables this is equivalent to $p_{\widetilde{x},\widetilde{y}}(x, y) = p_{\widetilde{x}}(x)p_{\widetilde{y}}(y)$ for every $x, y$.

**Independence implies uncorrelation.** With finite second moments, independence gives

$$\mathbb{E}[\widetilde{x}\widetilde{y}] = \mathbb{E}[\widetilde{x}]\mathbb{E}[\widetilde{y}] \quad \Longrightarrow \quad \mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0.$$

The reverse implication is false: covariance checks only one mixed moment, whereas independence constrains the entire joint distribution.

**Worked example: a nonlinear relationship with zero correlation.** Let $\widetilde{x}$ take $-1, 0, 1$ with equal probability, and set $\widetilde{y} = \widetilde{x}^2$. The possible pairs are

| $(x, y)$ | $(-1, 1)$ | $(0, 0)$ | $(1, 1)$ |
|---|---|---|---|
| $\mathbb{P}(\widetilde{x} = x, \widetilde{y} = y)$ | $1/3$ | $1/3$ | $1/3$ |

By symmetry,

$$\mathbb{E}[\widetilde{x}] = 0, \qquad \mathbb{E}[\widetilde{y}] = \frac{2}{3}, \qquad \mathbb{E}[\widetilde{x}\widetilde{y}] = \frac{-1 + 0 + 1}{3} = 0.$$

Hence $\mathrm{Cov}(\widetilde{x}, \widetilde{y}) = 0$, and $\rho_{\widetilde{x},\widetilde{y}} = 0$ because both variables have positive variance. However,

$$\mathbb{P}(\widetilde{x} = 0, \widetilde{y} = 0) = \frac{1}{3} \neq \frac{1}{9} = \mathbb{P}(\widetilde{x} = 0)\mathbb{P}(\widetilde{y} = 0).$$

They are dependent: observing $\widetilde{x}$ determines $\widetilde{y}$ exactly. The positive and negative contributions to covariance cancel.

**The Gaussian exception.** If $(\widetilde{x}, \widetilde{y})$ is *jointly Gaussian*, then uncorrelated $\Longleftrightarrow$ independent. It is not enough that each variable separately has a Gaussian distribution.

## 3 Review: correlation versus causation

Association describes how variables vary together. A causal question asks what would happen if we *intervened* to change one variable. For example, rain can cause both umbrella use and wet sidewalks:

$$\text{Umbrella use} \longleftarrow \text{Rain} \longrightarrow \text{Wet sidewalks}.$$

Their association does not establish that using an umbrella makes a sidewalk wet. A common cause is a **confounder**; reverse causation can also produce association. Randomized assignment can support a causal comparison. Observational adjustment requires additional causal assumptions, as in Recitation 5. Neither a large $|\rho|$ nor a small $|\rho|$ alone determines a causal effect; "variance explained" is a predictive statement.

## Practice problems

Three core problems; Problem 4 is a backup for a fast class. The indicated times are for independent work before discussion.

### Problem 1: Covariance properties (Exercise 8.2)

*Independent work: 10 minutes.* Let $\widetilde{a}$ and $\widetilde{b}$ have finite variances. Prove the following:

(a) For any constants $\alpha, \beta \in \mathbb{R}$,

$$\operatorname{Cov}(\beta\widetilde{a} + \alpha, \widetilde{b}) = \beta \operatorname{Cov}(\widetilde{a}, \widetilde{b}).$$

Conclude that if $\widetilde{a}$ and $\widetilde{b}$ are uncorrelated, so are $\beta\widetilde{a} + \alpha$ and $\widetilde{b}$.

(b) The covariance satisfies

$$-\sqrt{\operatorname{Var}(\widetilde{a})\operatorname{Var}(\widetilde{b})} \le \operatorname{Cov}(\widetilde{a}, \widetilde{b}) \le \sqrt{\operatorname{Var}(\widetilde{a})\operatorname{Var}(\widetilde{b})}.$$

Include the case where one variance is zero.

(c) The sum and difference satisfy

$$\operatorname{Cov}(\widetilde{a} + \widetilde{b}, \widetilde{a} - \widetilde{b}) = \operatorname{Var}(\widetilde{a}) - \operatorname{Var}(\widetilde{b}).$$

**Solution.** Define centered random variables $\widetilde{a}_0 = \widetilde{a} - \mathbb{E}[\widetilde{a}]$ and $\widetilde{b}_0 = \widetilde{b} - \mathbb{E}[\widetilde{b}]$.

**(a)** Since $\beta\widetilde{a} + \alpha - \mathbb{E}[\beta\widetilde{a} + \alpha] = \beta\widetilde{a}_0$,

$$\operatorname{Cov}(\beta\widetilde{a} + \alpha, \widetilde{b}) = \mathbb{E}[\beta\widetilde{a}_0\widetilde{b}_0] = \beta \operatorname{Cov}(\widetilde{a}, \widetilde{b}).$$

If the original covariance is zero, the transformed covariance is zero, including when $\beta = 0$.

**(b)** Cauchy–Schwarz gives

$$|\operatorname{Cov}(\widetilde{a}, \widetilde{b})| = |\mathbb{E}[\widetilde{a}_0\widetilde{b}_0]| \le \sqrt{\mathbb{E}[\widetilde{a}_0^2]\mathbb{E}[\widetilde{b}_0^2]} = \sqrt{\operatorname{Var}(\widetilde{a})\operatorname{Var}(\widetilde{b})}.$$

This proves both bounds. If a variance is zero, its centered variable is zero with probability one, so the covariance is also zero. The bound still holds, although correlation is then undefined.

**Alternative proof of (b).** If $\operatorname{Var}(\widetilde{b}) > 0$, set $t = \operatorname{Cov}(\widetilde{a}, \widetilde{b})/\operatorname{Var}(\widetilde{b})$. Then

$$0 \le \mathbb{E}[(\widetilde{a}_0 - t\widetilde{b}_0)^2] = \operatorname{Var}(\widetilde{a}) - \frac{\operatorname{Cov}(\widetilde{a}, \widetilde{b})^2}{\operatorname{Var}(\widetilde{b})}.$$

Rearrange and take square roots; handle zero variance as above.

**(c)** Center the sum and difference, then multiply:

$$\begin{aligned}\operatorname{Cov}(\widetilde{a} + \widetilde{b}, \widetilde{a} - \widetilde{b}) &= \mathbb{E}[(\widetilde{a}_0 + \widetilde{b}_0)(\widetilde{a}_0 - \widetilde{b}_0)] \\ &= \mathbb{E}[\widetilde{a}_0^2 - \widetilde{b}_0^2] = \boxed{\operatorname{Var}(\widetilde{a}) - \operatorname{Var}(\widetilde{b})}.\end{aligned}$$

No independence assumption is needed. Equal variances make the sum and difference uncorrelated; independence does not follow in general.

## Problem 2: An affine function (Exercise 8.11)

*Independent work: 7 minutes.* The random variables $\widetilde{a}$ and $\widetilde{b}$ satisfy

$$\mu_{\widetilde{a}} = 2, \qquad \mu_{\widetilde{b}} = -2, \qquad \sigma^2_{\widetilde{a}} = 4, \qquad \sigma^2_{\widetilde{b}} = 9, \qquad \rho_{\widetilde{a},\widetilde{b}} = 0.25.$$

Define $\widetilde{c} = 2\widetilde{b} + 3$. Find the linear minimum mean-squared-error predictor of $\widetilde{c}$ given $\widetilde{a} = a$.

*Suggested steps:* Convert correlation to covariance; find $\mathbb{E}[\widetilde{c}]$ and $\mathrm{Cov}(\widetilde{a}, \widetilde{c})$; then identify the slope and intercept of the predictor.

**Solution.** The standard deviations are $\sigma_{\widetilde{a}} = 2$ and $\sigma_{\widetilde{b}} = 3$, so

$$\mathrm{Cov}(\widetilde{a}, \widetilde{b}) = \rho_{\widetilde{a},\widetilde{b}} \sigma_{\widetilde{a}} \sigma_{\widetilde{b}} = \frac{1}{4} \cdot 2 \cdot 3 = \frac{3}{2}.$$

Linearity of expectation and the covariance rule from Problem 1 give

$$\mu_{\widetilde{c}} = 2(-2) + 3 = -1, \qquad \mathrm{Cov}(\widetilde{a}, \widetilde{c}) = 2\,\mathrm{Cov}(\widetilde{a}, \widetilde{b}) = 3.$$

Consequently,

$$\boxed{\ell_{\mathrm{MMSE}}(a) = \mu_{\widetilde{c}} + \frac{\mathrm{Cov}(\widetilde{a}, \widetilde{c})}{\mathrm{Var}(\widetilde{a})}(a - \mu_{\widetilde{a}}) = -1 + \frac{3}{4}(a - 2) = \frac{3}{4}a - \frac{5}{2}.}$$

**Check by transforming the predictor of $\widetilde{b}$.** First predict $\widetilde{b}$ by $-2 + \frac{3}{8}(a - 2)$. Multiplying this prediction by 2 and adding 3 gives the same answer. The predictor passes through the point $(\mu_{\widetilde{a}}, \mu_{\widetilde{c}}) = (2, -1)$.

**Optional discussion: how much does the prediction improve?** Here $\mathrm{Var}(\widetilde{c}) = 2^2 \cdot 9 = 36$, and the positive rescaling of $\widetilde{b}$ leaves the correlation unchanged: $\rho_{\widetilde{a},\widetilde{c}} = 1/4$. Therefore

$$\mathrm{MSE}_{\mathrm{linear}} = \left(1 - \frac{1}{16}\right) 36 = \frac{135}{4} = 33.75.$$

The constant predictor $-1$ has MSE 36, so the affine predictor explains $1/16 = 6.25\%$ of the variance. This is optional; the original exercise only asks for the predictor.

**Teaching point.** Use standard deviations to recover covariance from correlation, but use the input variance $\mathrm{Var}(\widetilde{a})$ in the slope denominator. The answer minimizes MSE over affine functions. It need not equal $\mathbb{E}[\widetilde{c} \mid \widetilde{a} = a]$; the exercise does not assume a jointly Gaussian distribution or an affine conditional mean.

### Problem 3: Ice cream sales (Exercise 8.15)

*Independent work: 8 minutes.* Alma is analyzing data for an ice cream company. The marketing team argues that a strong correlation between advertising expenses and sales shows that its advertising is effective. Alma obtains the following *sample covariance matrix*:

| | Advertising | Sales | Temperature |
|---|---|---|---|
| Advertising | 900 | 960 | 540 |
| Sales | 960 | 1600 | 720 |
| Temperature | 540 | 720 | 400 |

(a) Compute the sample correlation between advertising expenses and sales.

(b) Should Alma accept the team's conclusion? Use the temperature entries to inform your explanation, and distinguish association from a causal effect.

*Reminder:* Divide a sample covariance by the product of the sample standard deviations. The diagonal entries above are variances.

**Solution.** **(a)** The sample standard deviations of advertising, sales, and temperature are $\sqrt{900} = 30$, $\sqrt{1600} = 40$, and $\sqrt{400} = 20$. Thus

$$\boxed{\text{Sample correlation of advertising and sales} = \frac{960}{30 \cdot 40} = 0.8.}$$

This is a strong positive linear association in the observed data.

**(b)** Both quantities are even more strongly associated with temperature:

$$\text{Advertising–temperature correlation} = \frac{540}{30 \cdot 20} = 0.9,$$
$$\text{Sales–temperature correlation} = \frac{720}{40 \cdot 20} = 0.9.$$

A plausible explanation is that hotter weather increases demand for ice cream, and the company also spends more on advertising in hotter weather. Temperature could therefore be a common cause of advertising and sales:

$$\text{Advertising} \longleftarrow \text{Temperature} \longrightarrow \text{Sales}.$$

This is a possible causal explanation, not something established by the covariance matrix itself.

Alma can agree that advertising and sales are positively associated, but should **not accept the claim that the association proves the ads increase sales**. The matrix alone does not determine whether the causal effect of advertising is positive, zero, or negative.

**Discussion: what evidence would help?** Randomly assign different advertising levels to comparable markets or time periods, so assignment is independent of the potential sales outcomes, and compare sales. Avoid spillovers between experimental groups. With observational data, adjusting for temperature may help, but a causal interpretation additionally requires assumptions such as no remaining confounding and adequate overlap.

**Common mistakes.** The denominator is $30 \cdot 40$, not $900 \cdot 1600$. The value 0.8 does not mean that ads cause 80% of sales. Its square, 0.64, describes the sample variance explained by a simple OLS regression with an intercept, not the fraction of sales caused by ads. Also, identifying a plausible confounder does not prove that ads have no effect.

### Problem 4 (backup): A point in a disk

*Independent work: 5 minutes; use only if the class is ahead.* Choose a point uniformly from the unit disk. Its coordinates have joint pdf

$$f_{\widetilde{x},\widetilde{y}}(x,y) = \begin{cases} 1/\pi, & x^2+y^2 \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Without evaluating integrals, use symmetry to find $\mathbb{E}[\widetilde{x}]$, $\mathbb{E}[\widetilde{y}]$, and $\mathbb{E}[\widetilde{x}\widetilde{y}]$. Are the coordinates uncorrelated?

(b) Are the coordinates independent? Justify your answer using the disk's support or a pair of events.

**Solution.** **(a)** Reflecting the disk across the vertical axis preserves the distribution and changes the sign of $\widetilde{x}$, so $\mathbb{E}[\widetilde{x}] = 0$. Reflection across the horizontal axis similarly gives $\mathbb{E}[\widetilde{y}] = 0$. Reflecting across either axis reverses the sign of the product, hence $\mathbb{E}[\widetilde{x}\widetilde{y}] = 0$. Therefore

$$\boxed{\operatorname{Cov}(\widetilde{x},\widetilde{y}) = 0, \qquad \rho_{\widetilde{x},\widetilde{y}} = 0.}$$

Both variances are finite and positive because the coordinates are bounded and nonconstant.

**(b)** Let $A = \{\widetilde{x} > 3/4\}$ and $B = \{\widetilde{y} > 3/4\}$. Each event has positive probability: each corresponds to a region of positive area inside the disk. But if both occurred, then

$$\widetilde{x}^{\,2} + \widetilde{y}^{\,2} > \frac{9}{16} + \frac{9}{16} = \frac{9}{8} > 1,$$

which is impossible. Thus

$$\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B).$$

The coordinates are **dependent but uncorrelated**. Geometrically, once one coordinate is close to the edge of the disk, the other must be close to zero.

*Source:* adapted from the 2025 archive, `Recitations/Recitation 1 Problems.pdf`, Problem 2 [3.31]. The covariance question has been added.

## Instructor pacing guide (75 minutes)

| Time | Activity | Suggested use |
|---|---|---|
| 0–10 min | Correlation review | Definition, properties, affine prediction |
| 10–20 min | Uncorrelation vs. independence | Teach the worked counterexample |
| 20–25 min | Correlation vs. causation | Common causes; intervention |
| 25–42 min | Exercise 8.2 | 10 min individual work, 7 min discussion |
| 42–55 min | Exercise 8.11 | 7 min individual work, 6 min discussion |
| 55–71 min | Exercise 8.15 | 8 min individual work, 8 min discussion |
| 71–75 min | Wrap-up | Revisit the two invalid implications |

If the class finishes early, use Problem 4 with five minutes of independent work followed by discussion. Keep it optional; do not shorten the three core work periods to fit it in.

**Board plan for the mini-lecture.** Write "independent ⇒ uncorrelated" and prove it using factorization of the product expectation. Then ask whether the reverse holds. Work through the three-point example on page 2: list the pairs, compute the covariance, and disprove independence with one event pair. Emphasize that $\widetilde{y} = \widetilde{x}^2$ is perfectly predictable even though its best affine predictor is constant. For a numerical check, $\mathrm{Var}(\widetilde{x}) = 2/3$ and $\mathrm{Var}(\widetilde{y}) = 2/9$.

**Why the Gaussian exception needs "jointly."** For a jointly Gaussian pair, zero covariance makes the joint density factor into the product of its marginals (when both variances are positive). If needed, explain why Gaussian marginals alone are insufficient: take $\widetilde{z} \sim \mathcal{N}(0,1)$ and an independent fair sign $\widetilde{s} \in \{-1, 1\}$. Then $\widetilde{z}$ and $\widetilde{s}\widetilde{z}$ are each standard Gaussian and have covariance zero, but their absolute values are identical. They are dependent and are not jointly Gaussian.

**A short derivation of the affine predictor, if needed.** For a predictor $\alpha + \beta\widetilde{x}$, centering gives

$$\begin{aligned}\mathbb{E}[(\widetilde{y} - \alpha - \beta\widetilde{x})^2] = (\mu_{\widetilde{y}} - \alpha - \beta\mu_{\widetilde{x}})^2 + \sigma_{\widetilde{y}}^2 \\ - 2\beta\,\mathrm{Cov}(\widetilde{x}, \widetilde{y}) + \beta^2\sigma_{\widetilde{x}}^2.\end{aligned}$$

First set $\alpha = \mu_{\widetilde{y}} - \beta\mu_{\widetilde{x}}$; then minimize the quadratic in $\beta$, giving $\beta = \mathrm{Cov}(\widetilde{x}, \widetilde{y})/\sigma_{\widetilde{x}}^2$. Substitution gives the MSE formula on page 1.

**Closing check.** Ask students to explain why each statement can fail: "Zero correlation means independence" and "High correlation proves a causal effect." A nonlinear relationship addresses the first; a common cause addresses the second. Ask what extra assumption makes the first true and what study design can help with the second.

**Sources and adaptations.** The review follows Carlos Fernandez-Granda's *Probability and Statistics for Data Science*, Sections 8.2–8.6 and 8.8. Problems 1–3 rephrase Exercises 8.2, 8.11, and 8.15, retaining all their numerical data and requested conclusions; scaffolding has been added. The backup reuses the disk-independence question from the 2025 archive and adds a covariance calculation by symmetry. Its source handout is dated January 31, 2025.